ANSI/IES LM-41-20(R2023)

Illuminating
ENGINEERING SOCIETY

APPROVED METHOD:
PHOTOMETRIC TESTING OF INDOOR FLUORESCENT LUMINAIRES

AN AMERICAN NATIONAL STANDARD

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has been approved by IES.
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should be directed to IES.

Prepared by the
IES Testing Procedures Committee

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1.0 Introduction and Scope

1.1 Introduction

This guide is intended to provide uniform guidance useful for evaluating the performance of fluorescent luminaires for general lighting. Luminaire characteristics are to be reported in terms of intensity distribution, flux distribution and efficiency. These performance data may be developed into factors and/or tables that allow the predetermination of illuminance and luminance for any conventional plane or boundary surface.

1.2 Scope

This guide provides adequate and uniform methods for determining and reporting the photometric characteristics of indoor fluorescent luminaires. It describes characteristics of luminaires and some components, as well as the requirements for the thermal environment and proper control of the electrical and mechanical systems involved in testing. This document is also concerned with general test conditions and the testing procedure best suited for achieving accurate and consistent photometric results.

2.0 Normative References

2.1 ANSI C78.375A-2014

American National Standards Institute. American National Standard for Electric Lamps – Fluorescent Lamps – Guide for Electrical Measurements. New York: National Electrical Manufacturers Association; 2014.

2.2 ANSI C78.81-2016

American National Standard for Electric Lamps – Double-Capped Fluorescent Lamps – Dimensional and Electrical Characteristics. New York: National Electrical Manufacturers Association; 2016.

2.3 ANSI C82.3-2016

American National Standards Institute. American National Standard Specification for Electric Lamps – Reference Ballasts for Fluorescent Lamps. New York: National Electrical Manufacturers Association; 2016.

2.4 ANSI/IES LS-1-20

Illuminating Engineering Society. Lighting Science: Nomenclature and Definitions for Illuminating Engineering. New York: IES; 2020.

2.5 ANSI/IES LM-9-20/R23

Illuminating Engineering Society. Approved Method: Electrical and Photometric Measurements of Fluorescent Lamps. New York: IES: 2020.

2.6 ANSI/IES LM-54-20

Illuminating Engineering Society. Approved Method: Guide to Lamp Seasoning. New York: IES; 2020.

3.0 Definitions

The units of electrical measurement used in this approved method are the volt, the ampere, and the watt. The units of photometric measurement are the lumen and the candela (see ANSI/IES LS-1-20; see **Section 2**). Color is specified in terms of CIE recommended systems.

The industry-accepted point for making an initial rating of a discharge lamp occurs after the lamp's first 100 hours of operation per ANSI C78.81-2016 (see **Section 2**).

The cold chamber or cold spot is the location inside a fluorescent lamp with the lowest operating temperature. Both the location and temperature of the cold spot are critical for optimizing lamp performance.

4.0 Ambient and Physical Conditions

4.1 General

The photometric values and electrical characteristics of fluorescent lamps and luminaires are sensitive to ambient temperature and air movement.

4.2 Temperature

The ambient temperature shall be measured at a point not more than 1 m from the luminaire and at the same height as the luminaire. Under standard conditions this temperature shall be maintained at 25 °C ± 1 °C. In addition, the temperature sensor shall be shielded from the luminaire and radiation from any other source. Measurements at temperatures other than 25 °C are considered nonstandard and shall be so noted in the test report.

4.3 Air Movement

Air movement at the surface of a fluorescent lamp or luminaire under test may substantially alter electrical and photometric values. Therefore, no discernible airflow from sources other than the device under test shall be allowed. This can be tested using single-ply tissue paper hanging in place of the lamp; there should be no movement of the tissue.

5.0 Electrical Conditions and Instrumentation

5.1 Wave Shape

The AC power source during operation of the test lamp ballast or luminaire shall have a sinusoidal voltage wave shape such that the root-mean-square (RMS) summation of the harmonic components does not exceed 3.0% of the fundamental.

5.2 Voltage Regulation

The RMS voltage applied to the lamp ballast for the duration of the measurement interval shall be regulated to within ±0.1% for line voltage measurements and ±1% for frequencies above standard line frequency.

5.3 Power Source Impedance

The power source impedance for low frequency supplies, measured at the points where the ballast and lamp are connected, shall not exceed 2% of the ballast impedance. For high frequency power supplies it is not possible to meet the 2% limit internally in the power supply; therefore, electronic control circuits are used to keep the output voltage at the desired level. To ensure compliance with this requirement, variable autotransformers or other voltage transforming devices should have a kVA rating of at least five times the lamp wattage.

5.4 Instrumentation for Electrical Measurements

The instruments shall be calibrated at the manufacturer's recommended time interval, or at an interval determined from a documented statistical evaluation of the calibration stability. They shall have good reproducibility of indication. AC meters shall respond to true RMS.

Most electronic ballasts operate at a high frequency lamp current (typically greater than 20,000 Hz). Care should be taken that possible conduction or radiation from this type of ballast does not cause reading errors. High frequency leads should be as short as possible and, if necessary, shielded from electromagnetic interference (EMI). It may be necessary to use precautions in order to prevent EMI from reaching voltage monitoring equipment or the photocell readout device through the power circuits.

5.4.1. Uncertainties. The calibration uncertainties (see note below) of the instruments for AC voltage and current shall be ≤0.2%. The calibration uncertainty of the AC power meter shall be ≤0.5%, and that for DC voltage and current shall be ≤0.1%.

Note: *Uncertainty* here refers to relative expanded uncertainty with a 95% confidence interval, normally with a coverage factor k=2.[1,2] If a manufacturer's specification does not specify uncertainty this way, then the manufacturer should be contacted for proper conversion.

6.0 Test requirements

6.1 Selection and Preparation of Luminaire

The selection of sample luminaires is important because the value of the test will depend upon the method of sampling. If sampling is performed by the laboratory, the method used shall be reported.

Luminaires shall be selected to represent the purpose of the test, and shall be clean and free of any defect that could adversely affect the test results. Ballasts regularly furnished as part of the luminaire should be used to operate the lamps during the test and should be mounted in their normal location(s) within the luminaire. These ballasts should be within tolerances stated in the current version of *American National Standard Specification for Fluorescent Lamp Ballasts*,[3] when applicable. The ballast factor of the ballast(s) installed in the luminaire can significantly affect the photometric performance of the luminaire (see **Section 7.4**).

6.2 Luminaire Luminance Characteristics

Luminaire luminance characteristics are reported in order to facilitate evaluation of certain quality aspects of lighting design. Average luminance is a quantity calculated from photometric data (see **Section 7.6**).

6.3 Test Lamps

Lamps of stable output are needed for photometric purposes. They should be constant in light output during a test or test series when constant line voltage is supplied. The following points provide guidance for their selection and handling:

- Lamps should be typical of current production and should appear uniform in brightness and color from end to end and around the axis.
- Lamps for multi-lamp luminaires should be matched for light output within ±1.5% (a spread of 3%) when operated on the same supply and ballast circuit. The minor differences that occur between test lamps are sufficient reason for always photometering both sides of the luminaire.
- If the intensity distribution of a lamp in a plane at 90° to its long axis is designed to be other than circular, it should be checked to make certain that the shape of the intensity distribution is representative of its type. A circular distribution should be within ±2% (a 4% spread) of uniform. Readings should be taken in at least four locations, 90° apart, around the longitudinal lamp axis, and preferably, the entire 360° distribution should be scanned.
- The luminaire being checked for compliance should be mounted in a vertical or horizontal position with a mounting that does not shadow the lamp, or that can be altered after mounting to remove any shadowing. Usual care should be exercised in allowing for stray light corrections to obtain net relative values. The resultant uniformity of intensity distribution, U, is determined using:

  $$U = \frac{2(I_{max} - I_{min})}{I_{max} + I_{min}}$$, expressed as a percentage,

  where:

  U = Uniformity expressed as percent spread

  I_{max} = Relative net maximum reading observed during 360° scan

  I_{min} = Relative net minimum reading observed during 360° scan

  If U exceeds 4%, the lamp should be rejected.
- If the lamp type is one involving controlled mercury pressure regions, it is required that at all times (in storage, handling and test) it be kept in the position used during seasoning. Some compact fluorescent lamps and other nonlinear lamps may require an operating position other than horizontal for best results. The lamp manufacturer should be consulted for the recommended way to handle these sources. All lamps should be carefully protected against bumps or jolts during handling. Lack of such care may disturb the distribution of mercury in the tube and necessitate re-stabilization.

6.3.1 Special Considerations for Luminaires with Compact Fluorescent Lamps (CFLs). For the many specific and special conditions for the testing of compact fluorescent lamps, the reader is referred to the information in **Annex B**.

6.3.2 Lamp Transfer. Because of the time required for pre-burning, it is usually desirable to pre-burn lamps at a separate location from the photometric equipment. In this case, the test lamp is extinguished at the end of the pre-burning period and then transferred to the test location while maintaining the lamp orientation. As the test lamp is moved from one location to the other, it is important to keep it in the same physical orientation as maintained during pre-burning. Care should be taken to avoid shaking or bumping the lamp during transfer, as this could cause mercury to dislodge from the cool zones.

6.3.3 Lamp Seasoning. Prior to using lamps for photometric testing, they shall be seasoned per *ANSI/IES LM-54-20, Approved Method: Guide to Lamp Seasoning* (see **Section 2**).

6.4 Photometric Equipment

6.4.1 The Goniophotometry System. The complete goniophotometric apparatus and test range should be calibrated through the entire usable scale so that its individual readings can be traced to a national laboratory standard. Angular settings shall be reproducible within 0.1°.

6.4.2 Photometer. The goniophotometer should be designed so that the test luminaire can be correctly mounted (see **Section 7.5.1**) in relation to the goniophotometer optical axes. The goniophotometer should also provide the capability for determining luminous intensity and necessary angular settings in the test planes of the luminaire (see **Section 7.5.4**). In all cases, the mounting apparatus should not interfere with light emitted by the luminaire.

6.4.3 Automated Goniophotometer. A goniophotometric test system may be partially or fully automated through a computer. The linearity of the complete system should be maintained with frequent checks throughout its range.

6.4.4 Photometer Head (Photodetector). The photometer head of the goniophotometer should have relative spectral responsivity matched to the V(λ) function. The f_1' value of the spectral responsivity shall be less than 3%. For absolute measurements on spectrally structured sources (e.g., RGB or monochromatic), a systematic spectral correction factor should be considered. To determine such a correction factor, the spectral responsivity of the photometer head and the spectral distribution of the source to be measured should be known. It is important to note that there are photometer heads readily available that have an f_1' value for the spectral responsivity of less than 1.5%.

For determining total luminous flux of a luminaire or lamp using a goniophotometer, the photometer head shall have good cosine response within the angle range where light is incident, with the value of $f_{2\varepsilon,\phi}$ (relative deviation from the cosine function) less than 2% within the acceptance angle range. This good cosine response is required to in order accurately measure the candela values needed to calculate total luminous flux.

6.5 Luminaire Mounting

The luminaires shall be mounted under geometric conditions as nearly typical of end use as possible. For example, most luminaires for general interior lighting should be mounted in a horizontal position.

6.6 Lamp Circuit Connections

Lamp circuit pin connections shall remain the same throughout the test, including the calibration procedure.[4]

6.7 Stray Light

Provisions shall be made for eliminating stray light from other sources or reflections, i.e., any light that reaches the photometer head other than directly from the source to be measured. The presence of stray light can be detected by blocking off the direct light from the luminaire. The field of view of the photometer head should be limited (e.g., using aperture screens) in order to shield stray light reflected from the angles other than from the light source being measured. To minimize reflected stray light within the field of view of the photometer head, low-reflectance materials (such as black carpet, flat black paint, black velvet) are recommended for the wall and floor surfaces.

Any remaining stray light can be measured by running a complete test with the direct light from the luminaire completely shielded from the photometer head. This light can then be subtracted from the data, taking into account the variations of stray light for each vertical angle in each plane measured.

6.8 Luminaire or Lamp – Pre-burning and Stabilization

6.8.1 Pre-burning. *Pre-burning* refers to the operational warm-up of a luminaire or test lamp for the purpose of allowing the mercury in the lamp(s) to locate at the coolest zone of the lamp(s). Because the distribution of mercury within the discharge tube significantly affects electrical and photometric characteristics of fluorescent lamps, the pre-burning period is necessary to obtain repeatable measurements. Pre-burning also allows the ballast(s) and air temperature within the luminaire to reach thermal equilibrium, which is essential for stable light output. Pre-burning of the luminaire or test lamp shall be done in the same position that is to be used for the photometric test. It is recommended that luminaires or lamps be pre-burned for a minimum of 5 hours in the orientation in which they will be tested, and an overnight pre-burn is preferred. It is important to note that the pre-burning durations mentioned above are recommendations only; careful monitoring of stabilization as detailed in **Section 6.8.2** will allow testing to commence sooner if stabilization is achieved.

6.8.2 Stabilization. Stabilization immediately follows pre-burning and is the burning of test lamps for an additional period such that electrical and photometric values are constant. Stabilizing lamps prior to measurements is essential if results are to be reproducible in subsequent tests or are to be compared with measurements from other laboratories. Once pre-burning is complete and the lamps or luminaire have been quickly transferred warm to the test position, the lamps shall be burned long enough to obtain stabilization. If the test sample is being physically transferred from pre-burning to the test apparatus, extreme care should be taken to ensure that the sample is not jostled or the orientation of the sample changed (even momentarily) during the transfer. It can be judged that stability has been reached when the variation (the difference between maximum and minimum) of at least 3 readings of the light output and electrical power over a period of 30 minutes, taken 15 minutes apart, is less than 0.5%. For 800-ma or 1,500-ma lamps or for compact fluorescent lamps, stability is reached when at least 5 readings of the light output and electrical power over a period of 60 minutes, taken 15 minutes apart, is less than 0.5%. To ensure that stabilization has been achieved, these light output and electrical power measurements should not be increasing or decreasing monotonically.

7.0 Photometric Test Procedures

7.1 Test Distance

Photometric test distance is defined as the distance of the light path (direct and/or reflected) from the photometric center of the luminaire to the surface of the photoreceptor.[5-8]

The test distance is associated with the luminous intensity assigned to the luminaire. At any distance of sufficient length, the luminaire will have a theoretical luminous intensity that is equal to the square of the test distance times the illuminance produced. If the test distance is at least five times the maximum luminous dimension of the source, which in many cases is the diagonal dimension, the apparent luminous intensity will generally be within 1% of the theoretical point-source luminous intensity. This "five-times rule" is based on a 1% intensity error on the axis of a circular, Lambertian disk. It is well known that this distance may be inadequate for some interior luminaires. This often is true for interior luminaires with reflectors, lenses, or louvers that produce high intensity gradients and sharp cutoffs.[18] A series of readings at a longer test distance will confirm whether the longer distance is required. The actual test distance shall be reported in all cases.

For maximum accuracy, the test distance should be measured from the photodetector to the photometric center of the luminaire. The photometric center can be determined experimentally,[9] but this is a time consuming process. Furthermore, the photometric center may change with the angle of measurement (e.g., from louvers to lamp with a louvered luminaire). For simplicity and consistency, the luminaire should be mounted on the photometer so that the referenced plane (or part of the luminaire) coincides with the photometer axes as shown in **Table 7-1**.

7.2 Position Sensitivity

All fluorescent luminaires are sensitive to some degree to the influence of mounting position. Therefore, it is recommended that photometric testing be conducted with the lamp or luminaire operating in its normal end-use orientation. Where photometric testing is conducted with the luminaire mounted in other than its normal end-

Table 7-1. Luminaire Mounting Positions on Goniometer

Type of Luminaire	Photometric Center of Test Subject*
Recessed luminaires	To opening in major ceiling plane
Suspended luminaires	
• Up or down light only	To plane of major light opening
• Up and down light	To the photometric center of the luminaire
• Otherwise (lamps partially or wholly outside the major opening)	To geometric center of luminaire
Surface mounted luminaires	
• Direct classification	To plane of major light opening
• Indirect classification	To geometric center of lamp(s)

* *Table note*: To be positioned at the center of rotation of the goniophotometer

use position, the effect should be measured and data corrected to normal position values.[10] This is not usually practical with luminaires using extra-high output lamps, or U-shape, compact fluorescent, or circline lamps. In these cases, changing lamp position can change the location of the mercury pressure control regions, thus requiring extremely long stabilization time or producing an unstable condition. Even the most stable linear type fluorescent lamps are subject to changes in light output characteristics when experiencing a position change such as from horizontal to vertical.

Many luminaires are constructed in such a way that they should be photometered only in their normal mounting position. This precaution may be necessary to avoid mechanical disturbances because of sag, warping, or shift in position of critical light controlling components, or to avoid temperature variations due to changes in air flow.

7.3 Calibration of Photometer – Relative and Absolute Photometry

Until fairly recently (prior to 2006), most luminaires were tested with a goniophotometer using a procedure called *relative photometry*. When solid-state lighting (SSL)—specifically, LEDs—started making major lighting industry inroads around 2006, *absolute photometry* became more prevalent. Relative photometry cannot be used for SSL products because, in most SSL products, the LEDs are not designed to be separated from the luminaire. Even if the LED source can be separated and measured separately, the relative photometry method will not work accurately because the light output of the LED will change significantly if operated outside the luminaire, due to differences in thermal conditions. Therefore, relative photometric measurement of luminaires cannot be used for SSL products.

With solid state lighting generating such attention in the lighting industry, there has been increasing interest in performing absolute photometry on incumbent light sources, such as fluorescent, high intensity discharge (HID), and incandescent.

7.3.1 Relative Photometry – Calibration Factor. In relative photometry, calibration is necessary to relate the light output of the lamps used in the test to the lamp manufacturer's lumen rating in order to establish luminaire luminous intensity. The basic procedure is to determine the relative bare lamp lumens and to establish a calibration factor for the photometer with the particular lamp tested, using the relationship:

Calibration factor = (rated lamp lumens)/ (relative bare lamp lumens)

The lamp lumens can be measured in an integrating sphere. They may also be determined by measuring the intensity on a goniophotometer in a direction normal to the axis of a straight cylindrical lamp with an axially symmetric output. The equation to determine lumens from the intensity measurement is:

$$\Phi = K I_{\perp} ,$$

where:

Φ = luminous flux, lumens

$I_{\perp}$ = luminous intensity perpendicular to the lamp axis, candelas

K = lumens-to-luminous intensity ratio

And K is given by:

$$K = \frac{9.15}{\frac{D}{L}\tan^{-1}\left(\frac{L}{2D}\right) + \left(\frac{2D^2}{4D^2 + L^2}\right)} ,$$

where:

D = Test distance, meters

L = Lamp length, meters

Arctangent ($\tan^{-1}$) is measured in radians

The factor K is a function of both light distribution and a ratio of test distance to lamp length,[7,10,11] as shown in **Figure 7-1**. At a typical test distance of five times the lamp length, K is 9.21. Because of the difficulty in photometering bare lamps, it is preferable that the value of K from **Figure 7-1** be used (see also **Annex A**). For lamps to which **Figure 7-1** does not apply, it is necessary to determine the value through photometry of the bare lamps.

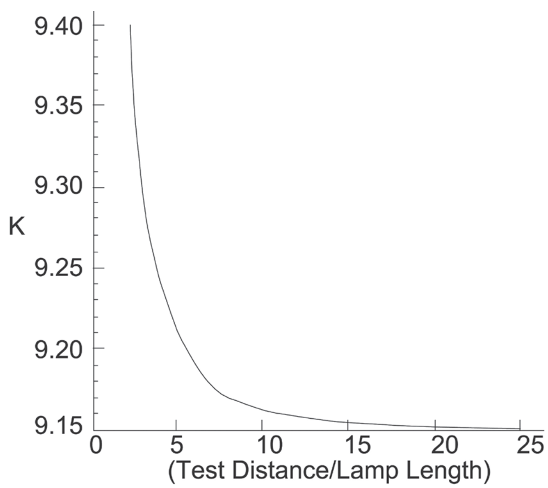

Figure 7-1. The factor K for axially symmetric linear fluorescent lamps, shown as a function of the ratio of test distance to lamp length.

If the lumens-to-luminous intensity ratio (K) is to be determined by complete photometry, it is important to maintain the highest degree of accuracy possible to avoid propagation of errors to the luminaire photometry.

The light output of very high output lamps, and U-shape, compact, and circline lamps is sensitive to orientation, ambient temperature fluctuations, and drafts, which puts considerable burden upon the photometric facility. The determination of K or luminous intensity normal to the longitudinal axis of the lamp, $I_{\perp}$, should be done without moving any of these lamp types. Inversion to allow viewing of the top and bottom of the lamps from the same position, for example, can give erroneous results.

Frequently, the goniophotometer construction will be such that it is not possible to view the entire lamp distribution without obstruction. In these cases extrapolation should be used to fill in that portion of the lamp distribution that cannot be measured without obstruction. The region over which such extrapolation is applied should be kept to the minimum possible; in no case should the region of extrapolation exceed 30° of cone (for example, extrapolation of the region 150° to 180° is acceptable; extrapolation of the region 140 to 180° is not acceptable). (See **Section 6.3** regarding lamp output symmetry.)

7.3.2 Relative Photometry – The Calibration Process. In relative photometry, the *calibration factor* is the value that will be used to multiply the relative readings obtained when the luminaire is measured. The result will be luminous intensity based on the assumed lumens of the lamps being tested. If more than one lamp is involved, the calibration factor will be an average.

The test distance, photodetector, electrical measuring instruments, line voltage, and basic procedure should be the same as for the luminaire test.

All lamps that operate from a single ballast should be operating during calibration even if the lamps are calibrated by separate exposure to the photodetector. The mated lamps should be operated in the same ambient temperature condition. Care should be exercised to ensure that inadvertent modification of the bulb wall temperature does not occur.

Lamps and ballasts should reach stable operating conditions before photometric readings are taken. This may be determined by observing when electrical and light readings become constant (see **Section 6.8.2**). If photometer calibration immediately follows a luminaire distribution test and the luminaire is already stable, less time will be required to reach a stable operating condition with the bare lamps (see **Section 6.8.2** for recommendations for minimum stabilization time and procedure).

Test lamps shall be mounted in a draft-free environment at 25 °C ± 1 °C (77 °F ± 2 °F) in the same orientation as when installed in the luminaire. A nonreflecting supporting frame should be so designed that it will minimize any changes in the temperature of the lamps.

The measured values will be simple relative instrument readings. The reading(s) should be taken perpendicular to the major axis of the lamp(s).

When photometry is being performed on luminaires with more than one lamp, the bare lamps may be measured singly (summing their readings) or in a group. If measured in a group, they should be mounted so as to prevent interreflections between lamps and mutual heating effects.[5] Care should be taken that baffles do not cause heat build-up.

Bare-lamp lumen ratings should be taken from the current lamp schedule for the particular manufacturer's lamps used in the test.

7.3.3 Relative Photometry – Special Considerations When Testing Lamps That Produce Maximum Light Output in an Ambient Temperature Other Than 25 °C. Fluorescent testing using the guidelines of relative photometric practice presupposes that the lamps will be operated at their nominal electrical characteristics (e.g., a 40-watt lamp will operate very nearly at 40 watts, and at the voltage and current required for 40-watt operation). Fluorescent lamps in general are temperature sensitive, and their lumen output varies with ambient temperature, following a characteristic curve. When using T5 fluorescent lamps or other lamps that produce maximum light output in an ambient temperature other than 25 °C, certain challenges exist. T5 lamps used for this example peak in light output at 35 °C. A critical step in relative photometric testing involves measurement of the total light output from the lamp(s) suspended in free air at an ambient temperature of 25 °C, per **Section 4.2**. This measurement process is a separate step from the photometric exploration of the luminaire itself. This "bare lamp" measurement is made with the lamp(s) operated by the same ballast(s) that are to be used in the luminaire. Since the test procedure involves measuring the "bare lamp" light output at 25 °C, and T5 lamps used for this example peak at a temperature of 35 °C, the "bare lamp" total flux measured for this lamp type will be less than the maximum output the lamp is designed to produce. When this "bare lamp" measurement is incorporated into the luminaire test report, the net effect is that total luminaire efficiency is higher than what the lighting industry historically would expect this luminaire to produce. These lighting industry expectations are based on comparisons to the total luminaire efficiency of the same luminaire with T8 or T12 lamps. In addition, if the lamp lumen rating at 35 °C was used to generate the report instead of the lumen rating at 25 °C, the candela and lumen values on the luminaire test report will be higher than what the luminaire actually produced.

7.3.4 Absolute Photometry. Absolute photometry measures the light output of the luminaire under specified operating conditions directly with reference to a calibrated measurement standard, with an assigned responsivity and with stated uncertainties.

This type of measurement requires careful evaluation of the responsivity of the goniophotometer's photodetector-amplifier system.

With this method, it is necessary to obtain an absolute factor (*k*) for the photometer using a luminous intensity (candela) standard lamp traceable to national standards, a total luminous flux (lumen) standard lamp traceable to national standards, or by directly comparing the goniophotometer's detector system to a detector calibrated for illuminance responsivity traceable to national standards.

It is suggested but not required that the absolute factor (*k*) be established using one of the traceable calibration standard pathways and then validated within the goniophotometer measurement uncertainty budget with one of the other traceable pathways.

7.4 Ballast Factor

7.4.1 Ballast Factor Defined. *Ballast factor* is a term used to describe the fraction of flux output produced when a fluorescent lamp is energized from a commercially available ballast, as compared to the light output produced when energized from a *reference ballast*. (See **Section 2.3 ANSI C82.3-2016**, **Section 2.4 LS-1-20**, and **Informative Reference 12**.)

A standard reactor ballast may be used as a reference ballast. However, it may be more convenient to use a production ballast with a known ballast factor as a *calibrated ballast*. Production ballasts with a measured ballast factor are available from most ballast manufacturers. They are also available from several independent photometric laboratories. A production ballast used as a calibrated ballast should have a ballast factor measured in accordance with ANSI C82.3-2016 (see **Section 2**). The calibrated ballast should be rated for the same lamps to be used with the test ballast.

7.4.2 Determination of Ballast Factor. The bare lamp should be mounted in accordance with **Section 7.3.2** and operated on a warm-up ballast until its light output is stabilized. The lamp is then switched to the calibrated ballast and a reading taken within 30 seconds. The lamp is then switched back to the warm-up ballast. The test ballast is substituted for the calibrated ballast, the lamp is switched to the test ballast, and a second reading is taken within 30 seconds. The lamp should not be extinguished during any of the switching. This procedure applies to ballasts operating any number of lamps.

The ballast factor of the test ballast is determined by:

$$BF_t = BF_c \times (LO_t/LO_c)\ ,$$

where:

BF_t = ballast factor of the test ballast

BF_c = ballast factor of the calibrated ballast

LO_t = light output with the test ballast

LO_c = light output with the calibrated ballast

7.4.3 Effect of Ballast Factors on Photometric Test Results. Since fluorescent luminaires are typically tested using relative photometry, the data will be reported as if the lamp(s) produced rated lumens. The candela and lumen data shown on the report will be correct for a ballast factor of 1.0. However, the candela and lumen data shown on the report will be higher than what is actually produced when ballasts with low ballasts factors are used, and lower than what is actually produced when ballasts with high ballasts factors are used. That is, for a ballast with a ballast factor of 0.80, the candela and lumen data shown on the report will be 25% higher than what the luminaire actually produces [(reported data)/(ballast factor) = 1.0/0.80 = 1.25, or 25% high]; conversely, for a ballast with a ballast factor of 1.25, the candela and lumen data shown on the report will be 20% lower than what the luminaire actually produces [(reported data)/(ballast factor) = 1.0/1.25 = 0.80, or 20% low].

The light output of fluorescent lamps is affected by ambient temperature. Consequently, the ballast factor of the ballast in a luminaire affects the lamp output directly by electrical control and indirectly by changing the lamp power and thus the ambient temperature within the luminaire. Therefore, the ballast factor of the ballast used to drive the lamps plays a major role in the absolute amount of light that a luminaire with fluorescent lamps will produce. Two examples are provided.

Example A: Luminaires that are designed to have many lamps in a small air-tight lamp cavity will likely produce more light using a ballast having a lower ballast factor (less than 1.0), than if it was equipped with a ballast having a ballast factor of 1.0 or greater. This is contrary to what one might believe since ballasts with higher ballast factors should drive the lamps harder and should produce higher light output. However, with many lamps in a tight lamp cavity, the temperature within the lamp cavity exceeds the optimum temperature at which the fluorescent lamps are designed to be operated, and hence the lamps produce less light.

Example B: Luminaires that are designed to have just a few lamps in large open lamp cavities will likely produce more light when using a ballast having a higher ballast factor of 1.0 or greater. This occurs because the temperature within the large open lamp cavity is closer to the optimum temperature at which the fluorescent lamps are designed to be operated when driven by ballasts with higher ballast factors.

7.4.4 Application of Ballast Factors. It is recommended but not required to include the ballast factor(s) for the ballast-lamp combination(s) used for the photometric test as part of the test report. It should be noted that with relative photometry, the tabulated intensities, curves, flux, and CU values, which are given in the report, should all be calculated assuming a ballast factor of 1.0.

7.5 Luminaire Photometry

7.5.1 Mounting. The luminaire should be mounted on a photometer that is designed for minimum obstruction of light. Care should be exercised to prevent light from reflecting off of the mounting apparatus directly into the photodetector, or back onto the luminaire where the photodetector might pick it up.

7.5.2 Position of Luminaire during Test. Photometric readings should be taken with the luminaire in normal end-use orientation (see also **Section 7.2**).

7.5.3 Angles for Measurement. The number of photometric measurement points in various vertical planes and the angular spacing between these measurement points should be such as will permit plotting of well defined luminous intensity distribution curves. Steps at 2.5-degree intervals in vertical planes are usually satisfactory. Shorter intervals are recommended where the luminous intensity from the luminaire is changing rapidly or is erratic, such as with luminaires using parabolic louvers or specular faceted reflectors.

7.5.4 Planes for Measurement. The number of planes reported should be determined by the nature of the distribution as to symmetry or irregularity and the end result desired from the test. For luminaires whose intensity has quadrilateral symmetry, the five reporting planes [parallel (0°), 22.5°, 45° 67.5°, and normal (90°) to the lamp axis] are usually adequate. For bilaterally symmetric distributions, it will be necessary to report measurement planes from 0° to 180°. In extreme cases (luminaires with rapidly varying lateral intensity), it may be necessary to reduce the lateral step size to well below 22.5° in order to adequately capture the intensity gradient.

Because of variations in test lamp performance, luminaire fabrication, and the forming techniques of luminaire shielding, luminaire raw data measurements should be made in all four quadrants, at whatever lateral step size (see previous paragraph) is deemed adequate. The raw data measurements should be averaged so as to present the desired reporting angles according to the symmetry of the distribution. For example, for quadrilateral symmetry, the intensities at plane 22.5° would be averaged with those at 157.5°, 202.5°, and 337.5°.

Lateral photometric angles shall be reported in a counterclockwise direction, viewing the luminaire from above.

Tabulated luminous intensity values are preferred on the report, as they provide the most accurate information. If luminous intensity distribution curves are plotted on the test report for any reason other than to simply show distribution shape, they should be clearly defined.

In the case of a luminaire with bilaterally symmetric distribution (e.g., a wall washer), it is necessary that the luminaire orientation should be clearly indicated with respect to the beam position. For electronic published data, it is recommended that 0° be on the beam side.

7.5.5 Polarization Measurements. For measuring the percent polarization of light from a luminaire at each photometric angle, one should follow the general rules provided in **References 13** and **14**. Means should be provided to make the photodetector sensitive to light in a given plane of polarization, i.e., light defined as either horizontally or vertically polarized light as defined in **Reference 13**. Since filter devices such as linear polarizing material have appreciable light absorption characteristics, it is necessary to measure both the horizontal and vertical components of polarization to obtain the percentage of polarization of light in either plane. It should be noted that any photometric test range that employs mirrors may introduce errors in polarization measurements because of the slightly polarizing characteristics of mirrors themselves.

7.6 Luminance Measurements

The average luminance of the luminaire should be calculated for planes perpendicular, at 45°, and parallel to the lamp axis at the following vertical angles from nadir: 0°, 45°, 55°, 65°, 75° and 85°. The value at 0° is seldom required. Average luminance is a quantity calculated from photometric data. It is given by the luminous intensity at a given angle divided by the projected area of the luminous surface at that angle in square meters or square feet.[15]

7.7 Computation of Luminous Intensity Distribution Test Results

The photometric readings should be converted into luminous intensity based on lamps theoretically delivering rated lumens in open air by applying the multiplication factor developed by calibration of the photometer with the test lamps, as referred to in **Section 7.3**.

This establishes luminous intensity distribution and makes possible subsequent calculations of zonal lumens and luminous efficiency.[16]

7.8 Computing Zonal Lumens

Zonal lumens may be established for each angular zone by a weighted average of the luminous intensity values in each of the planes of measurement and multiplying by the appropriate zonal constant from **Annex A**. For five planes, the 0° and 90° planes are taken once, and the 22.5°, 45°, and 67.5° planes twice each. The sum is then divided by 8.

7.9 Overall Luminaire Efficiency

Luminaire efficiency may be computed by dividing the total emitted lumens (sum of zonal lumens) by the total rated lumen output of the lamps. For the purpose of illumination design, the percentage of upward component (lumens emitted in the 90° to 180° zone) and the corresponding percentage of downward component should be given. The sum is the overall efficiency.[16]

7.10 Coefficients of Utilization

Coefficients of utilization shall be calculated using the recommended IES calculation procedure.[17]

8.0 Test Reports

It is suggested that test report forms follow IES recommended types.[18] Test reports should describe all significant data concerning the luminaire tested, and the data should be recorded in a manner that will give at least the following information:

a. Luminaire description
 1. Manufacturer's name
 2. Type and catalog number of luminaire
 3. Type of mounting
 4. Reflectance of painted surfaces, if available
 5. Description of enclosing media, including louver surface finish, texture, and specularity
 6. Bare lamp shielding angle measured down from horizontal, both parallel and normal to the lamp axis[3]
 7. Sketch showing luminaire shape, dimensions, and light center position(s)
 8. Ballast data: catalog number of ballast(s), and any additional pertinent information related to the ballast
 9. Input watts

b. Test lamp description
 1. Number, type, bulb designations, and length where applicable
 2. Rated watts
 3. Rated lumen output
 4. Color

c. General lighting characteristics
 1. Luminous intensity distribution curves in designated planes
 2. Tabulated luminous intensity values for each plane at measured angles, or in angular steps that fine enough to be deemed useful in interpreting and using the data)
 3. Tabulations of lumens and percentages of bare lamp lumens for zones 0° to 30°, 0° to 40°, 0° to 60°, 0° to 90°, 90° to 180° and 0° to 180° (for totally indirect luminaires, tabulation should be in the 90° to 120°, 90° to 130°, 90° to 150°, and 90° to 180° zones)
 4. Curves or tabulations of average luminance data
 5. Test distance

d. Date of test and testing agency

e. Description

A complete description of any item or procedure peculiar to the luminaire being tested and/or its report (e.g., if the luminaire returns air through the lamp cavity or side air slots, it should be noted whether the door, top, or side slots are opened or closed). If there is no provision to close slots, they should be noted as open.

f. Statement

Include this statement: "This photometric test was performed using a specific ballast/lamp combination. Extrapolation of these data for other ballast/lamp combinations may produce erroneous results."

g. Statement

Include a statement to the effect that the ballast factor should be applied to the test data for all calculations. For example: "*NOTE*: This photometric test was conducted in accordance with test procedure *ANSI/IES LM-41-20* in which the ballast(s) and lamp(s) are presumed to produce 100% rated output. The ballast factor shown on this test report should be applied as a correction factor to the lumen output rating assigned to the lamps or, correspondingly, to the luminous intensity values given."

h. Statement

Include this statement: "This photometric test was conducted in controlled laboratory conditions; ambient temperature was held at 25 °C + 1 °C (77 °F ± 2 °F). Field performance may differ, particularly in regards to change in luminaire light output as a result of difference in ambient temperature and the way in which a luminaire is mounted."

Note: Statements described in bullets **f**, **g**, and **h** are suggested so as to warn the user of particular factors involved in producing a photometric test report that may change under actual applied conditions. Equivalent statements may be used individually or in combination to best suit the needs of the agency preparing the test report.

9.0 ADDITIONAL READING

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Annex A – Zonal Lumen Constants

Formula for determining zonal constants (see also **Table A-1**):

$\phi_n = K_\omega I$,

where:

ϕ_n = flux in zone n

I = 360° average about polar axis of intensity at center of zone n

K_ω = weighting factor, equal to solid angle of zone n:

$K_\omega = 2\ \pi(\cos\ \theta_1 - \cos\ \theta_2)$,

where:

θ_1 = lower angular limit of zone n

θ_2 = upper angular limit of zone n

Annex B – Special Considerations for Compact Fluorescent Lamps (CFLs)

The single-ended compact fluorescent lamp is a low-pressure mercury-discharge light source, with compact geometry achieved by using two or more folded arc tubes or two or more individual tubes connected by a bridge. Some designs include a “cold chamber,” which provides control of the mercury vapor pressure. (This excludes circline and U-bent lamps, which do not have specific cold chambers and do not have the pre-burning and stabilization requirements.)

As a general guide, consideration should be given to the burning position so that the mercury droplets remain in the cold spot in the “base up” or “base down” position.

Table A-1. ***K*** **Constants for Use in the Zonal Method of Computing Luminous Flux from Luminous Intensity Data**

5° Zones		10° Zones	
Zone Limits (degrees)	Zonal Constant	Zone Limits (degrees)	Zonal Constant
0 – 5	0.0239	0 – 10	0.095
5 – 10	0.0715	10 – 20	0.283
10 – 15	0.1186	20 – 30	0.463
15 – 20	0.1649	30 – 40	0.628
20 – 25	0.2097	40 – 50	0.774
25 – 30	0.2531	50 – 60	0.897
30 – 35	0.2946	60 – 70	0.993
35 – 40	0.3337	70 – 80	1.058
40 – 45	0.3703	80 – 90	1.091
45 – 50	0.4041		
50 – 55	0.4349		
55 – 60	0.4623		
60 – 65	0.4862		
65 – 70	0.5064		
70 – 75	0.5228		
75 – 80	0.5351		
80 – 85	0.5434		
85 – 90	0.5476		

The handling of compact fluorescent lamps for photometric measurements requires special care to reduce the time necessary for re-stabilization.[19]

Pre-burning refers to the burning of a test lamp for the purpose of allowing the mercury to locate at the coolest zone of the lamp. Because the distribution of mercury within the discharge tube significantly affects electrical and photometric characteristics of the lamp, the pre-burning period is necessary to obtain repeatable measurements.

Stabilization refers to the burning of test lamps for a sufficient period such that electrical and photometric values are constant. Stabilizing lamps prior to measurements is essential if results are to be reproducible in subsequent tests or are to be compared with measurements from other laboratories.

The actual temperature at which light output is at maximum will depend on the lamp design as well as lamp tilt and/or roll. The incidence of drafts on the surface of a lamp under test might substantially alter electrical and photometric values. Slight drafts that might easily escape normal observation can cause important changes in the measured values. When measurements of fluorescent lamp(s) are being made, extreme caution should be exercised to reduce air movement to a minimum if accurate and reproducible results are to be obtained.

A full scan of the lamp is required for the proper evaluation of photometer calibration when relative photometry is used. The distribution of luminous intensity around a lamp may be determined in a photometer similar to that used for normal luminous intensity measurements, but provided with a means for varying the angles between the detector and the lamp axis.

Since most single-ended compact fluorescent lamps require that all excess mercury be condensed in pockets or cooling chambers for proper stabilization, these lamps may have orientation altered during photometry, but tilt and roll should remain unaltered (refer to *ANSI/IES LM-72-20, Approved Method: Directional Positioning of Photometric Data*).[20]

It should be recognized that peak light output of some single-ended compact fluorescent lamps might not be achieved at ambient temperatures of 25 °C (77 °F). Some single-ended compact fluorescent lamps are designed with cold chambers consisting of tips that extend away from the path of the discharge. This provides a cooler place for the mercury to condense, lowering the mercury vapor pressure within the lamp and thus raising the ambient temperature value at which luminous output is maximum.

If there is any uncertainty as to whether the published nominal lumen value for a test lamp is the value for the standard reference temperature of 25 °C (77 °F), the lamp manufacturer should be contacted and the matter should be resolved before the test lamp is used (refer to **Section 2.5 IES LM-9-09(R17**) and **Informative Reference 20**). In the event that the value is not per the standard but is for an ambient temperature other than 25 °C (77 °F), a test for light output versus temperature will have to be run that includes 25 °C (77 °F) and the lamp ambient temperature at which lamp lumen output is maximum (or other non-standard published value). The ratio of the light output at these points is the correction factor to be used in determining the lamp lumen value at 25 °C (77 °F) and the calibration of the photometer.

B.1 Special Considerations for CFLs in a Non-base-up Orientation

There are widespread applications where lamps are burned base-down or horizontally. In the base-down position, the cooling tips or cold chambers will be located at the uppermost part of the lamp. As the lamp burns, mercury will migrate to the cooling tips just as it would if the lamp were burned base-up. However, in a base-down orientation, the slightest vibration or movement of the lamp will cause mercury droplets to fall to the bottom, or cathode end, of the discharge tube, raising the mercury vapor pressure and thus causing the luminous flux of the lamp to change. The test lamp will then tend to exhibit cyclical variations of light output and electrical values.

In actual applications, the instability of lamps burned base-down is not visually noticeable, but for photometric measurements, repeatability of results is very difficult to obtain. Therefore, base-down positioning is not recommended as a standard procedure.

For lamps having more than two discharge tubes, there will usually be cooling tips at both ends of the lamp. In the base-down position, the mercury will collect at the tips in the lowest part of the lamp, which are coolest. Therefore, stability can be achieved in the base-down position or the base-up position, provided that test lamps are handled according to the recommendations provided in this document. For base-down photometry, lamps should also be pre-burned base-down.

For lamps burned in the horizontal position, it is possible to achieve stability, but a much longer time is required. Therefore, as a standard method, burning only in the horizontal position without initial burning in the vertical position is not recommended. It is suggested that the lamp be first pre-burned base-up until stabilization is achieved. Then the test lamp can be moved slowly without vibration or burning to a horizontal position. Measurements can then be taken when a new stable lumen value is reached.

B.2 Special Considerations for Twin-Tube Fluorescent Lamps

Season the twin tube lamp for 100 hours in a horizontal position such that a plane passing through the central axis of both tubes is parallel to the floor. After seasoning, the lamps are maintained in the same horizontal position so as to minimize the mercury re-stabilization time. The lamps are mounted in the photometric facility in the same position as they will be used in the luminaire to be photometered. This procedure is to be done in draft-free air at 25 °C $\pm$ 1 °C (77 °F $\pm$ 2 °F), according to standard IES procedure. The lamps are operated on the commercial ballast(s) that will be used in the test luminaire for a minimum of 12 hours to ensure thermal stabilization. At rated line voltage, a light cell reading is obtained in a referenced direction for lamps for which a reference intensity is available, or a complete scan is run on the lamp if referencing to a nominal lumen value. Then the calibration factor is determined.

The lamps are maintained in the horizontal position when transported and when inserted into the test luminaire. At all times, the base of each lamp should be level with or higher than the opposite bend area of the lamp so that the mercury will remain in the cold spot location that has been established in the bend area. Unless the procedure is carefully followed, significant errors might be induced in measuring light output. Highly loaded lamps, such as those in the twin-tube family, require additional care to ensure stability of the cold spot during comparative photometric measurements.

When the lamps are inserted into the luminaire, they shall always be maintained in the horizontal orientation dictated by the specific geometry of the luminaire lamp holder.

The lens or louver shall be attached according to the luminaire manufacturer's instructions so that the luminaire is now ready for normal operation. The luminaire shall be operated at least 6 hours to ensure thermal stabilization of the ballast(s) and luminaire. The electrical as well as the photometric values shall be measured according to standard IES procedures.

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